Homework 14

For Monday, November 9, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, angles are measured in radians, and all definite integrals are Riemann integrals.

1. Use substitution to evaluate

$$\int_0^1 (1-2x)^4\,dx.$$

Show the transformed bounds.

2. Let $f:\mathbb{R}\to\mathbb{R}$ be continuous, with $\int_1^5 f(u)\,du = 7$. Determine

$$\int_0^2 x f(1+x^2)\,dx.$$

3. Evaluate

$$\int_0^{1/\sqrt{2}} \frac{x}{\sqrt{1-x^4}}\,dx.$$

4. Evaluate

$$\int_0^3 \frac{x}{\sqrt{x+1}}\,dx.$$

5. Evaluate

$$\int_0^{\pi/2} \frac{\sin x}{2+\cos x}\,dx.$$

6. Find the differentiable function $F:\mathbb{R}\to\mathbb{R}$ satisfying

$$F'(x) = (3x^2+1)e^{x^3+x}, \qquad F(0) = 2.$$

Verify your answer.

7. Find

$$\int \frac{1}{x\ln x}\,dx,$$

and state the maximal open intervals on which the integrand is defined and the formula is valid.

8. Construct two antiderivatives F, G of $1/x$ on $\mathbb{R} \setminus \{0\}$ such that $G - F$ is not constant on that whole domain. Explain why this does not contradict the theorem that antiderivatives differ by a constant on an interval.

9. Does

$$\int_0^2 \frac{1}{x-1}\,dx$$

exist as a Riemann integral? Explain why evaluating $[\ln|x-1|]_0^2$ does or does not give the value of this integral.

10. Let $A(x) = \int_0^x e^{t^2}\,dt$. Use substitution to prove

$$A(x) = x\int_0^1 e^{x^2u^2}\,du \qquad \text{for every } x \in \mathbb{R}.$$

Include the cases $x < 0$ and $x = 0$. Do not try to find an explicit antiderivative of e^{t^2}.

11. A particle has position $s(t)$ and velocity

$$s'(t) = (t-1)(t-3), \qquad 0 \le t \le 4,$$

with $s(0) = 2$. Find $s(4)$ and the total distance traveled between $t = 0$ and $t = 4$.

12. Evaluate

$$I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx.$$

Use the substitution $u = \pi/2 - x$ to obtain a second expression for I, then add the two expressions.

13. Use integration by parts to evaluate

$$\int_0^1 x\cos(\pi x)\,dx.$$

14. Use integration by parts to evaluate

$$\int_1^e \ln x\,dx.$$

You may write the integrand as $1 \cdot \ln x$.

15. Evaluate

$$\int_0^1 x^2 e^x \, dx$$

by applying integration by parts twice.

16. Let f be continuously differentiable on $[0, 1]$, with

$$f(1) = 3, \qquad \int_0^1 f(x) \, dx = 2.$$

Determine $\displaystyle\int_0^1 x f'(x) \, dx$.

17. Let u, v be continuously differentiable on $[a, b]$, with $a < b$, and let $C \in \mathbb{R}$. Prove that

$$[u(x)(v(x) + C)]_a^b - \int_a^b u'(x)(v(x) + C) \, dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, dx.$$

Explain what this says about the constant chosen when finding v from v' in integration by parts.

18. Let f have two continuous derivatives on $[0, 1]$, and suppose $f(0) = f(1) = 0$. Prove that

$$\int_0^1 f(x) f''(x) \, dx = -\int_0^1 (f'(x))^2 \, dx \leq 0.$$

Show that equality holds if and only if $f(x) = 0$ for every $x \in [0, 1]$. You may use the fact that a continuous nonnegative function with integral zero is identically zero.

19. Find the limit by recognizing a Riemann sum and then evaluating the corresponding integral:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \frac{2(k/n)}{1 + (k/n)^2}.$$

20. Find all real numbers b satisfying

$$\int_0^b (1 + x)e^x \, dx = 2e^2.$$

Prove that there are no other solutions.